Capacity Analysis

Viva revision guide — key concepts, formulas, and quick-fire Q&A

1. What is Capacity?

Capacity is the maximum rate of output of a process or facility over a given period. It is measured either in terms of *output* (units per hour) or *inputs* (machine-hours, labour-hours, seats, beds). Capacity decisions are long-term, capital-intensive, and difficult to reverse — they set the upper limit on what operations can deliver.

2. Types of Capacity

- **Design capacity**: the maximum theoretical output under ideal conditions.
- **Effective capacity**: maximum possible output given product mix, scheduling difficulties, maintenance, quality factors and breaks.
- **Actual output**: what is really produced — always ≤ effective capacity.
- **Rated / sustainable capacity**: realistic long-run output the firm can sustain.

3. Key Measures — Efficiency & Utilisation

```
Utilisation = Actual output / Design capacity
Efficiency = Actual output / Effective capacity
Expected output = Effective capacity × Efficiency
```

> **Mnemonic:** Utilisation compares to the *dream* (design); Efficiency compares to the *realistic plan* (effective). Utilisation is always ≤ Efficiency.

4. Determinants of Effective Capacity

- **Facilities**: size, layout, location, climate control.
- **Product / service**: design, standardisation, mix.
- **Process**: technology, automation, quality issues.
- **Human factors**: skills, motivation, training, absenteeism.
- **Policy**: shifts, overtime, work hours.
- **Operational**: scheduling, inventory, balancing of work.
- **Supply chain**: reliable suppliers, distribution.
- **External**: regulation, unions, pollution control.

5. Capacity Planning Horizons

Horizon	Time frame	Decisions
Long-term	> 1 year	Add plant, new technology, expansion
Medium-term	6–18 months	Hiring, sub-contracting, additional shifts
Short-term	Daily – weekly	Overtime, schedule changes, work transfers

6. Capacity Strategies

- **Lead strategy**: add capacity in anticipation of demand — captures market, but risks idle capacity.
- **Lag strategy**: add capacity only after demand is proven — conservative, but risks lost sales / dissatisfied customers.
- **Match (tracking) strategy**: add capacity in small increments to follow demand.
- **Capacity cushion** = 100% – Utilisation. Higher cushion suits volatile demand; lower cushion suits stable, capital-intensive industries.

7. Bottleneck Analysis & Theory of Constraints

The **bottleneck** is the resource with the lowest capacity in a process — it dictates the throughput of the entire system. Improving non-bottleneck resources does *not* increase output. Goldratt's **Theory of Constraints (TOC)** formalises this insight.

Five Focusing Steps of TOC

- **Identify** the system's constraint.
- **Exploit** the constraint (squeeze maximum output from it).
- **Subordinate** everything else to the constraint.
- **Elevate** the constraint (add capacity if needed).
- **Repeat** — once broken, a new constraint will appear; do not let inertia set in.

```
Throughput = min(capacity of each stage)
Cycle time = 1 / Throughput rate
```

8. Break-Even Analysis

Used to find the volume at which total revenue equals total cost — a key input to capacity sizing.

```
Break-even (units) Q* = FC / (P − VC)
Break-even (revenue) = FC / (1 − VC/P)
Profit = Q × (P − VC) − FC
```

FC = fixed cost, VC = variable cost per unit, P = price per unit. Larger capacity raises FC but typically lowers VC — the choice depends on expected volume.

9. Decision Tools

- **Decision trees**: structure capacity choices under uncertainty using Expected Monetary Value (EMV).
- **Waiting-line (queuing) analysis**: size service capacity to balance customer wait against staffing cost.
- **Simulation**: model variability where formulas don't apply.
- **Linear programming**: allocate limited capacity across products to maximise profit.
- **Learning curves**: forecast capacity gains from experience.

10. Economies & Diseconomies of Scale

- **Economies of scale**: average cost falls as output rises — fixed costs spread, bulk discounts, specialisation.
- **Diseconomies of scale**: average cost rises beyond a point — coordination overhead, complexity, fatigue, distribution costs.

- Each plant has an optimal operating level; **best operating level** is the volume at which average unit cost is minimised.

11. Worked Mini-Example

A bakery has a design capacity of 1,000 loaves/day. Effective capacity is 800 loaves/day after allowing for cleaning and changeover. Actual output is 720 loaves/day.

Utilisation = 720 / 1000 = **72%**
Efficiency = 720 / 800 = **90%**

If demand rises to 900 loaves/day, the bakery cannot meet it within effective capacity — options: overtime (short-term), extra shift (medium-term), or new oven (long-term).

12. Quick-Fire Viva Q&A

Q. Define capacity.

A. The maximum rate of output a process or facility can achieve over a given time, expressed in units of output or input.

Q. Difference between design and effective capacity?

A. Design is the ideal-conditions maximum. Effective capacity is what is realistically achievable given product mix, scheduling, maintenance and quality losses.

Q. Why is utilisation always ≤ efficiency?

A. Because design capacity ≥ effective capacity, and both ratios use actual output in the numerator. So dividing by the larger denominator gives a smaller fraction.

Q. What is a bottleneck and why does it matter?

A. The lowest-capacity stage in a process — it caps the throughput of the entire system, so improvements elsewhere have no effect on output.

Q. What is a capacity cushion?

A. The amount of capacity in excess of expected demand, i.e. 100% – utilisation. Used as a buffer against demand or supply variability.

Q. When would you choose a lead strategy over a lag strategy?

A. When growth is rapid and predictable, the cost of lost sales is high, or being first to market is strategically important.

Q. Explain the Theory of Constraints in one line.

A. A system's output is governed by its weakest link; focus improvement on the constraint and subordinate everything else to it.

Q. How do you do break-even capacity analysis?

A. Compute $Q^* = FC / (P - VC)$ for each capacity alternative; pick the option whose break-even volume sits comfortably below expected demand.

Q. What are economies and diseconomies of scale?

A. Economies: average cost falls as scale grows (fixed-cost spreading, bulk buying). Diseconomies: beyond an optimal point, coordination and complexity push average cost back up.

Q. How does product mix affect capacity?

A. Different products have different processing times. A richer mix or more setups reduces effective capacity even though design capacity is unchanged.

Q. What is the role of forecasting in capacity planning?

A. Capacity is committed long before demand is realised — forecasts of volume, mix and growth drive the size, timing and type of capacity added.

Q. How would you analyse capacity for a cafe?

A. Identify the bottleneck (counter / espresso machine / seating), measure throughput per hour, calculate utilisation during peak vs off-peak, and use queuing analysis to size staff and equipment for the peak demand.

Q. Why is capacity decision strategic?

A. It commits large capital, is hard to reverse, sets a ceiling on revenue, affects cost structure and competitive position.

One-Minute Recap

Capacity = max output. We distinguish **design** vs **effective** vs **actual**, and measure performance via **utilisation** and **efficiency**. Capacity is planned across **long, medium and short** horizons, using **lead, lag or match** strategies. Throughput is set by the **bottleneck** — the Theory of Constraints says: identify, exploit, subordinate, elevate, repeat. **Break-even analysis, decision trees and queuing models** support the choice. Watch out for the curve of **economies vs diseconomies** of scale.